

PrimeCore Minerals

Centralised Network Monitoring and Logging Implementation

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Date: September

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1. Introduction

This document provides an explanation of how the network monitoring and centralised logging for the PrimeCore Minerals Brisbane HQ were set up. The HQ-MONITOR server is used to collect and view network activity and pfSense security events. SPAN is used to capture network traffic, while pfSense logs are sent to the server and displayed through a Grafana dashboard.

The monitoring solution was later expanded so the Brisbane HQ monitoring server could also receive and display firewall events from Mine Site A. This allows the HQ-MONITOR server to act as a central location for viewing firewall and security activity from both locations.

By centralising the monitoring, it improved upon the project MVP, allowing events from both Brisbane HQ and Mine Site A to be viewed from the same Grafana dashboard. Authentication logging from the PRIMECORE.TEST domain was also added so successful and failed login attempts could be monitored alongside the firewall events.

2. Purpose

Monitoring and logging are part of the project MVP and are required to provide visibility into the activity taking place across the PrimeCore network.

By implementing the monitoring and logging setup, it provides the team with a central way to monitor activity across Brisbane HQ and Mine Site A. The monitoring solutions that have been used are an important component for my role as Security and Validation Lead, as they allow me to verify network traffic and security events during testing and ensure security controls are functioning as intended.

SPAN monitoring lets me view copied network traffic in a simple and easy way, while pfSense logging and Grafana provide a visual representation of firewall traffic and events.

3. Monitoring Design

The HQ-MONITOR server was added to Server VLAN 90 and uses two network interfaces for different monitoring tasks. I configured SPAN monitoring through Open vSwitch to copy network traffic to the monitoring server. SPAN works by copying traffic from a selected source and sending that copy to another interface for monitoring. The pfSense trunk port ens4 on the Open vSwitch was used as the source because it carries traffic between pfSense and the different VLANs.

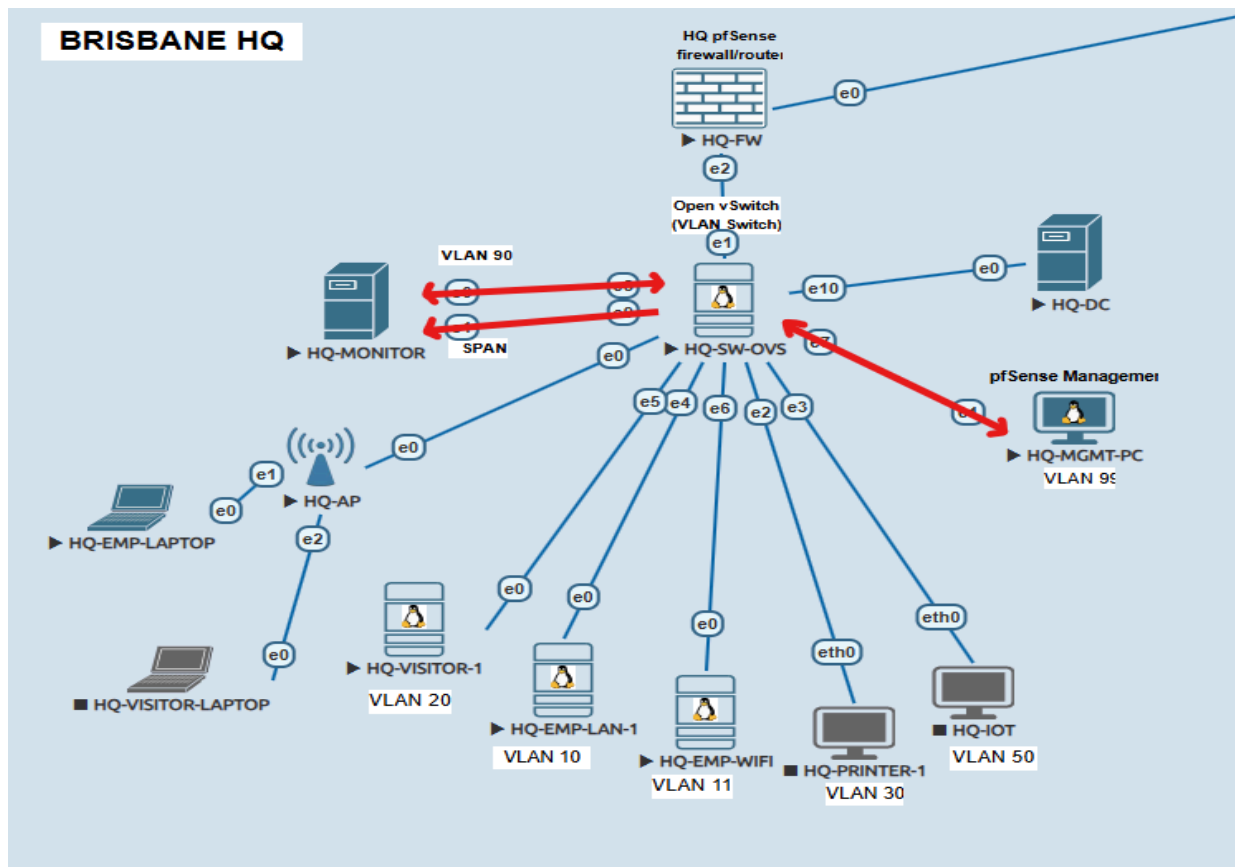


Figure 1: HQ monitoring design showing the separate SPAN and central logging connections

3.1 HQ-MONITOR Interface Configuration

Interface	Purpose	Configuration
ens3	Central logging and Grafana access	192.168.90.60/24
ens4	SPAN packet monitoring	None

3.2 Centralised Logging Sources

The central logging design uses HQ-MONITOR as the main collection point for more than one source. HQ-FW sends its pfSense events directly through Server VLAN 90, SITE-A-FW sends its firewall events to HQ-MONITOR through the site-to-site IPsec VPN, and HQ-DC provides authentication events from the PRIMECORE.TEST domain.

Source	Address	Path to HQ-MONITOR	Stored log / label
HQ-FW	192.168.90.1	VLAN 90 using UDP 514	/var/log/pfsense.log site="brisbane-hq"
SITE-A-FW	192.168.190.1	IPsec VPN using UDP 514	/var/log/sitea-pfsense.log site="mine-site-a"
HQ-DC	192.168.90.10	Central authentication logging	/var/log/samba-auth.log job="samba"

4. SPAN Network Monitoring

To make the activity on the Brisbane HQ network more accessible for viewing, I created a monitoring server and configured SPAN monitoring through Open vSwitch on it. SPAN works by creating a copy of traffic from a specific source and sending it to a specified device. The pfSense trunk interface 'ens4' was used as the source because it is the main connection carrying network traffic between pfSense and the different VLANs. The traffic then gets copied to ens12, which is the mirror output connected to HQ-MONITOR.

On HQ-SW-OVS:

```
sudo ovs-vsctl \  
-- --id=@src get Port ens4 \  
-- --id=@dst get Port ens12 \  
-- --id=@m create Mirror name=HQ-SPAN \  
select-src-port=@src \  
select-dst-port=@src \  
output-port=@dst \  
-- set Bridge br0 mirrors=@m
```

On HQ-MONITOR:

```
sudo tcpdump -eni ens4
```

The packet capture showed traffic from multiple VLANs and displayed the VLAN tags associated with the packets.

```

length 0
14:55:52.962292 50:00:00:01:00:02 > 00:50:00:00:07:00, ethertype 802.1Q (0x8100), length 101: vlan 10, p 0, ethertype IPv4 (0x0800), 104.20.23.154.443 > 192.168.10.72.43584: Flags [P.], seq 5624:5655, ack 1828, win 16, options [nop,nop,TS val 4196960388 ecr 3767957144], length 31
14:55:52.962431 00:50:00:00:07:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 101: vlan 10, p 0, ethertype IPv4 (0x0800), 192.168.10.72.43584 > 104.20.23.154.443: Flags [P.], seq 1828:1859, ack 5624, win 302, options [nop,nop,TS val 3767957148 ecr 4196960387], length 31
14:55:52.968093 50:00:00:01:00:02 > 00:50:00:00:07:00, ethertype 802.1Q (0x8100), length 229: vlan 10, p 0, ethertype IPv4 (0x0800), 104.20.23.154.443 > 192.168.10.72.43584: Flags [P.], seq 5655:5814, ack 1859, win 16, options [nop,nop,TS val 4196960394 ecr 3767957148], length 159
14:55:52.968566 00:50:00:00:07:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 70: vlan 10, p 0, ethertype IPv4 (0x0800), 192.168.10.72.43584 > 104.20.23.154.443: Flags [I.], ack 5814, win 314, options [nop,nop,TS val 3767957154 ecr 4196960388], length 0
14:55:52.970869 00:50:00:00:07:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 118: vlan 10, p 0, ethertype IPv4 (0x0800), 192.168.10.72.43584 > 104.20.23.154.443: Flags [P.], seq 1859:1907, ack 5814, win 314, options [nop,nop,TS val 3767957156 ecr 4196960388], length 48
14:55:52.970870 00:50:00:00:07:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 94: vlan 10, p 0, ethertype IPv4 (0x0800), 192.168.10.72.43584 > 104.20.23.154.443: Flags [P.], seq 1907:1931, ack 5814, win 314, options [nop,nop,TS val 3767957157 ecr 4196960388], length 24
14:55:52.971990 00:50:00:00:07:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 70: vlan 10, p 0, ethertype IPv4 (0x0800), 192.168.10.72.43584 > 104.20.23.154.443: Flags [F.], seq 1931, ack 5814, win 314, options [nop,nop,TS val 3767957157 ecr 4196960388], length 0
14:55:52.975273 50:00:00:01:00:02 > 00:50:00:00:07:00, ethertype 802.1Q (0x8100), length 70: vlan 10, p 0, ethertype IPv4 (0x0800), 104.20.23.154.443 > 192.168.10.72.43584: Flags [I.], ack 1931, win 16, options [nop,nop,TS val 4196960401 ecr 3767957156], length 0
14:55:52.976626 50:00:00:01:00:02 > 00:50:00:00:07:00, ethertype 802.1Q (0x8100), length 70: vlan 10, p 0, ethertype IPv4 (0x0800), 104.20.23.154.443 > 192.168.10.72.43584: Flags [F.], seq 5814, ack 1932, win 16, options [nop,nop,TS val 4196960402 ecr 3767957157], length 0
14:55:52.977029 00:50:00:00:07:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 70: vlan 10, p 0, ethertype IPv4 (0x0800), 192.168.10.72.43584 > 104.20.23.154.443: Flags [I.], ack 5815, win 314, options [nop,nop,TS val 3767957163 ecr 4196960402], length 0
14:55:57.582489 00:50:00:00:0b:00 > ff:ff:ff:ff:ff:ff, ethertype 802.1Q (0x8100), length 331: vlan 90, p 0, ethertype IPv4 (0x0800), 0.0.0.0.68 > 255.255.255.255.67: BOOTP/DHCP, Request from 00:50:00:00:0b:00, length 285
14:55:58.702738 00:50:00:00:0b:01 > ff:ff:ff:ff:ff:ff, ethertype IPv4 (0x0800), length 327: 0.0.0.0.68 > 255.255.255.255.67: BOOTP/DHCP, Request from 00:50:00:00:0b:01, length 285
14:56:40.509848 00:50:00:00:08:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 102: vlan 20, p 0, ethertype IPv4 (0x0800), 192.168.20.72 > 192.168.10.72: ICMP echo request, id 19332, seq 1, length 64
14:56:41.557943 00:50:00:00:08:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 102: vlan 20, p 0, ethertype IPv4 (0x0800), 192.168.20.72 > 192.168.10.72: ICMP echo request, id 19332, seq 2, length 64
14:56:42.581972 00:50:00:00:08:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 102: vlan 20, p 0, ethertype IPv4 (0x0800), 192.168.20.72 > 192.168.10.72: ICMP echo request, id 19332, seq 3, length 64
14:56:43.606040 00:50:00:00:08:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 102: vlan 20, p 0, ethertype IPv4 (0x0800), 192.168.20.72 > 192.168.10.72: ICMP echo request, id 19332, seq 4, length 64
14:56:45.910186 00:50:00:00:08:00 > 50:00:00:01:00:02, ethertype 802.1Q (0x8100), length 46: vlan 20, p 0, ethertype ARP (0x0806), Request who-has 192.168.20.1 tell 192.168.20.72, length 28
14:56:45.910861 50:00:00:01:00:02 > 00:50:00:00:08:00, ethertype 802.1Q (0x8100), length 46: vlan 20, p 0, ethertype ARP (0x0806), Reply 192.168.20.1 is-at 50:00:00:01:00:02, length 28

```

Figure 2: SPAN packet capture on HQ-MONITOR displaying copied traffic from trunk.

5. Centralised pfSense Logging

In addition to SPAN monitoring, both pfSense firewalls were configured to send security and system events to HQ-MONITOR using UDP port 514. rsyslog receives the events and stores the Brisbane HQ and Mine Site A logs in separate files so they can be identified and monitored independently.

5.1 Brisbane HQ pfSense Logging

The logs are sent through VLAN 90 on ens3 and HQ-MONITOR, which uses the static IP address 192.168.90.60/24. The dedicated rsyslog configuration was created on HQ-MONITOR using:

On HQ-MONITOR

```
sudo nano /etc/rsyslog.d/10-pfsense.conf
```

```
module(load="imudp")
input(type="imudp" port="514")

if ($fromhost-ip == '192.168.90.1') then {
    action(type="omfile" file="/var/log/pfsense.log")
    stop
}
```

This rsyslog configuration tells HQ-MONITOR to listen for syslog messages on UDP port 514. It verifies the messages are coming from pfSense at 192.168.90.1. If they are, the events are saved into /var/log/pfsense.log so the pfSense logs can be stored and reviewed from the monitoring server.

```

labadmin@srv-lnx-01:~$ sudo tail -f /var/log/pfsense.log
[sudo: authenticate] Password:
2026-09-11T06:09:15+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,17412,0,DF,17,udp,250,192.1
68.1.169,192.168.1.255,6537,6537,230
2026-09-11T06:09:15+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,16487,0,DF,17,udp,32,192.16
8.1.32,255.255.255.255,40706,10001,12
2026-09-11T06:09:15+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,17424,0,DF,17,udp,250,192.1
68.1.169,192.168.1.255,6537,6537,230
2026-09-11T06:09:15+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,6824,0,DF,17,udp,216,192.16
8.1.68,255.255.255.255,34206,6667,196
2026-09-11T06:09:16+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,0,0,DF,17,udp,248,192.168.1
.221,255.255.255.255,54286,6667,228
2026-09-11T06:09:17+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,255,34077,0,none,17,udp,216,19
2.168.1.27,255.255.255.255,49154,6667,196
2026-09-11T06:09:18+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,255,27646,0,none,17,udp,216,19
2.168.1.26,255.255.255.255,49154,6667,196
2026-09-11T06:09:19+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,255,29754,0,none,17,udp,216,19
2.168.1.28,255.255.255.255,49154,6667,196
2026-09-11T06:09:19+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,255,1554,0,none,17,udp,200,192
.168.1.99,255.255.255.255,65060,6667,180
2026-09-11T06:09:19+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,54280,0,DF,17,udp,200,192.1
68.1.229,255.255.255.255,49597,6667,180
2026-09-11T06:09:20+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,17750,0,DF,17,udp,250,192.1
68.1.169,192.168.1.255,6537,6537,230
2026-09-11T06:09:20+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,17773,0,DF,17,udp,250,192.1
68.1.169,192.168.1.255,6537,6537,230
2026-09-11T06:09:20+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,6825,0,DF,17,udp,216,192.16
8.1.68,255.255.255.255,34206,6667,196
2026-09-11T06:09:21+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,64,0,0,DF,17,udp,248,192.168.1
.221,255.255.255.255,54286,6667,228
2026-09-11T06:09:21+00:00 192.168.90.1 filterlog[62179]: 1,71,,12004,vtnet0,match,block,in,4,0x0,,255,34078,0,none,17,udp,216,19
2.168.1.27,255.255.255.255,49154,6667,196

```

Figure 3: The stored /var/log/pfsense.log pfSense events displayed on HQ-MONITOR

✗	Aug 26 13:39:13	WAN	192.168.1.229:39666	255.255.255.255:6667	UDP
✗	Aug 26 13:39:14	vtnet2	0.0.0.0:68	255.255.255.255:67	UDP
✗	Aug 26 13:39:14	WAN	192.168.1.99:65060	255.255.255.255:6667	UDP
✗	Aug 26 13:39:14	WAN	192.168.1.28:49154	255.255.255.255:6667	UDP
✗	Aug 26 13:39:15	WAN	192.168.1.221:54286	255.255.255.255:6667	UDP
✗	Aug 26 13:39:15	WAN	192.168.1.26:49154	255.255.255.255:6667	UDP
✗	Aug 26 13:39:15	EMPLOYEE_VLAN	0.0.0.0	192.168.10.1	ICMP
✗	Aug 26 13:39:15	WAN	192.168.1.32:57801	255.255.255.255:10001	UDP
✗	Aug 26 13:39:16	WAN	192.168.1.27:49154	255.255.255.255:6667	UDP
✗	Aug 26 13:39:17	EMPLOYEE_VLAN	0.0.0.0	192.168.10.1	ICMP
✗	Aug 26 13:39:18	WAN	192.168.1.229:39666	255.255.255.255:6667	UDP
✗	Aug 26 13:39:19	WAN	192.168.1.99:65060	255.255.255.255:6667	UDP
✗	Aug 26 13:39:19	WAN	192.168.1.28:49154	255.255.255.255:6667	UDP
✗	Aug 26 13:39:19	WAN	192.168.1.221:54286	255.255.255.255:6667	UDP
✗	Aug 26 13:39:20	WAN	192.168.1.26:49154	255.255.255.255:6667	UDP
✗	Aug 26 13:39:21	WAN	192.168.1.27:49154	255.255.255.255:6667	UDP
✗	Aug 26 13:39:23	WAN	192.168.1.229:39666	255.255.255.255:6667	UDP

Figure 4: pfSense firewall logs showing blocked network traffic.

5.2 Mine Site A pfSense Logging

SITE-A-FW uses 192.168.190.1 on the Site A Server VLAN. Its firewall events are sent to HQ-MONITOR at 192.168.90.60 using UDP port 514 through the site-to-site IPsec VPN. This keeps the Site A logging traffic inside the encrypted inter-site connection.

On HQ-MONITOR

```
sudo nano /etc/rsyslog.d/21-sitea-pfsense.conf
```

```
if ($fromhost-ip == "192.168.190.1") then {  
    action(type="omfile" file="/var/log/sitea-pfsense.log")  
    stop  
}
```

The Site A events are stored separately in /var/log/sitea-pfsense.log. Keeping the two firewall log files separate makes it easier to identify where an event came from while still allowing both locations to be viewed through the same monitoring platform.

5.3 Active Directory Authentication Logging

HQ-DC sends authentication events from the PRIMECORE.TEST domain to HQ-MONITOR so login activity can be monitored alongside the firewall logs. The events are stored in /var/log/samba-auth.log and then read by Grafana Alloy. This allows successful and failed domain login attempts to be viewed through Grafana.

6. Grafana Monitoring

Although the pfSense logs could already be viewed through the HQ-MONITOR terminal, it was not a permanent solution and was difficult to understand. To provide a more improved form of monitoring and logging, I installed Grafana, Loki and Grafana Alloy on HQ-MONITOR so the pfSense events could be viewed through a visual dashboard instead.

Tool	Function
Grafana Alloy	Reads the pfSense log file and forwards the events
Loki	Stores and searches the collected logs
rsyslog	Receives pfSense syslog events
Grafana	Displays the logs and other information through a custom web dashboard

To configure Grafana Alloy, I opened its configuration file using:

```
sudo nano /etc/alloy/config.alloy
```

I then added the following command:

```
loki.source.file "pfsense" {
  targets = [{
    __path__ = "/var/log/pfsense.log",
    job = "pfsense",
    site = "brisbane-hq",
  }]
  forward_to = [loki.write.local.receiver]
}

loki.source.file "samba" {
  targets = [{
    __path__ = "/var/log/samba-auth.log",
    job = "samba",
  }]
}
```

```
    site = "brisbane-hq",
  ]]
  forward_to = [loki.write.local.receiver]
}

loki.source.file "pfsense_sitea" {
  targets = [{
    __path__ = "/var/log/sitea-pfsense.log",
    job = "pfsense",
    site = "mine-site-a",
  }]
  forward_to = [loki.write.local.receiver]
}
```

This configuration tells Grafana Alloy to read the Brisbane HQ, Mine Site A and authentication log files and send the events into Loki. Labels are used to separate the different sources so they can be viewed individually through Grafana.

Grafana then gets accessed from HQ-MGMT-PC using the URL <http://192.168.90.60:3000>. Within Grafana I added Loki as the data source using <http://127.0.0.1:3100>. I used this address because Grafana and Loki are both running on the same HQ-MONITOR server, which allows Grafana to access the Loki logs locally.

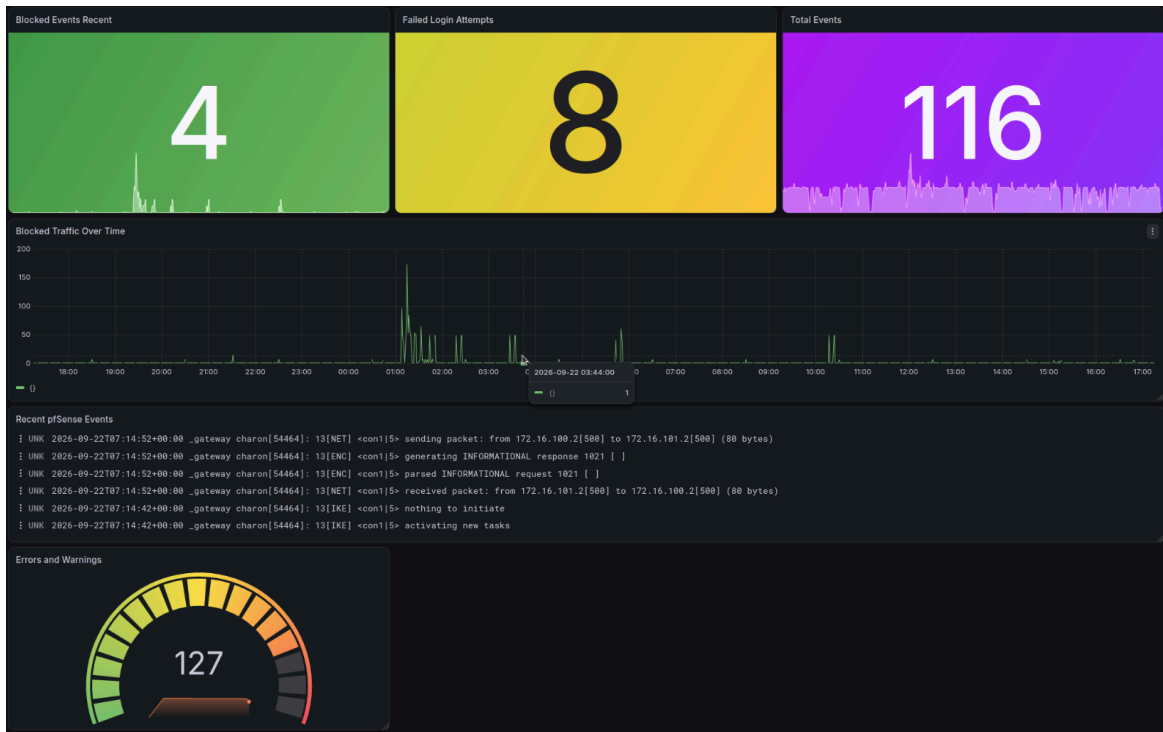


Figure 5: Custom Grafana dashboard displaying pfSense network and security events for Brisbane HQ.

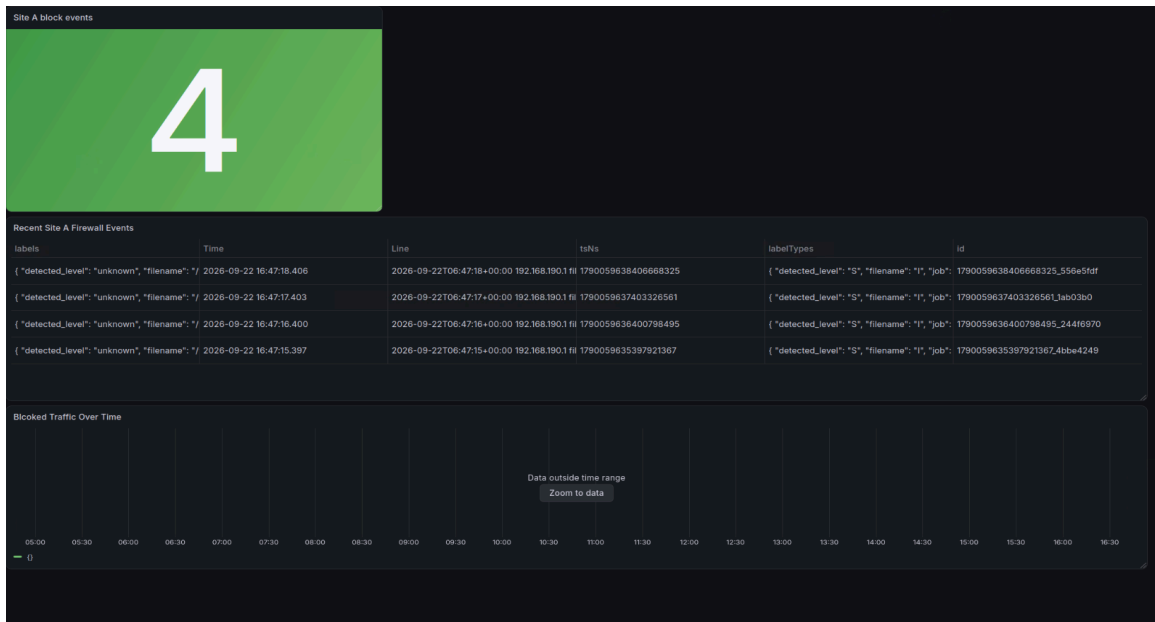


Figure 6: Grafana dashboard for Mine Site A to display pfSense network and security events.

Once Loki was connected to Grafana, I created a custom dashboard to make the pfSense logs easier to view and understand. Using queries, I was able to separate the different types of events and display them in panels.

Brisbane HQ Dashboard Queries	
Panel	Query code
Blocked Events Recent	<code>sum(count_over_time({job="pfsense", site="brisbane-hq"} = "block," [\$__range]))</code>
Total Events	<code>sum(count_over_time({job="pfsense", site="brisbane-hq"} [\$__range]))</code>
Recent pfSense Events	<code>{job="pfsense", site="brisbane-hq"}</code>
Blocked Traffic Over Time	<code>sum(count_over_time({job="pfsense", site="brisbane-hq"} = "block," [\$__interval]))</code>
Errors and Warnings	<code>sum(count_over_time({job="pfsense", site="brisbane-hq"} ~ "(?i)(error warning failed)" [\$__range]))</code>

Mine Site A Dashboard Queries	
Panel	Query code
Total Blocked Events	<code>sum(count_over_time({job="pfsense", site="mine-site-a"} = "block" [\$__range]))</code>
Recent pfSense Events	<code>{job="pfsense", site="mine-site-a"}</code>
Blocked Traffic Over Time	<code>sum(count_over_time({job="pfsense", site="mine-site-a"} = "block" [1m]))</code>

7. Monitoring Access and Security

The Grafana dashboard is being hosted from the HQ-MONITOR through the address 192.168.90.60 using TCP port 3000. To restrict access to only staff with administrator permissions, I configured pfSense so only the HQ-MGMT-PC on VLAN 99 can connect to the dashboard. This keeps the dashboard unavailable to general staff on Employee and Visitor networks.

Rules (Drag to Change Order)											
	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 2/76 KIB	IPv4 TCP	MANAGEMENT_VLAN subnets	*	This Firewall (self)	MANAGEMENT_PORTS	*	none		R10b -	
☐	✓ 0/0 B	IPv4 TCP	MANAGEMENT_VLAN subnets	*	This Firewall (self)	MANAGEMENT_PORTS	*	none		R10a - Allow Management Access to pfSense	
☐	✓ 0/16 KIB	IPv4 TCP/UDP	HQ_AP	*	*	DNS_PORT	*	none		HQ AP DNS	
☐	✓ 0/2.00 MIB	IPv4 TCP	HQ_AP	*	*	WEB_PORTS	*	none			
☐	✓ 0/0 B	IPv4 UDP	*	*	*	NTP_PORT	*	none			
☐	✓ 0/0 B	IPv4 TCP	192.168.99.71	*	192.168.90.60	3000 (HBCI)	*	none			

Figure 7: pfSense Management VLAN firewall rule allowing HQ-MGMT-PC on VLAN 99 to access the Grafana dashboard through port 3000.

8. Testing and Validation

Test	Expected Result	Actual Result	Status
SPAN capture	HQ-MONITOR should see copied traffic from the HQ VLANs.	Tagged traffic from multiple VLANs was visible in tcpdump.	PASS
HQ pfSense syslog	HQ-FW events should reach HQ-MONITOR.	HQ firewall events were received over UDP 514.	PASS
HQ log storage	HQ events should be saved in /var/log/pfsense.log	Events were written to the expected file.	PASS
Site A firewall logging	SITE-A-FW events should reach HQ-MONITOR through the VPN.	Site A firewall events were received over UDP 514.	PASS
Site A log storage	Site A events should be saved in /var/log/sitea-pfsense.log	Events were written to the Site A log file.	PASS
Alloy to Loki	Alloy should read the configured log files and send them to Loki.	Firewall and authentication sources were available in Loki	PASS
Grafana pfSense dashboard	Firewall events should be visible through Grafana.	HQ firewall events were displayed in the dashboard.	PASS
Site A Grafana monitoring	Site A events should be separated using the mine-site-a label.	Site A events and blocked traffic were displayed.	PASS
Samba authentication logging	HQ-DC authentication events should be collected.	Authentication events were available through the Samba source.	PASS
Failed login monitoring	An incorrect domain password should create a visible failed login event.	The failed GuestUser login appeared in Grafana.	PASS

Management access	HQ-MGMT-PC should open Grafana on TCP 3000.	The dashboard was accessible from HQ-MGMT-PC	PASS
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9. Security, Risk and Change Management

To provide security access to the Grafana dashboard was limited to HQ-MGMT-PC using a pfSense firewall rule. This ensures Employee and Visitor devices are prevented from accessing the monitoring system and reduces the risk of the dashboard configuration being changed or viewed by unauthorised people connected to the network. The logging setup also keeps Brisbane HQ, Mine Site A and authentication events separated using different log files and labels, which makes it easier to identify where security events have come from. When improving the logging and monitoring solution, I kept the original rsyslog configuration working and added Grafana, Loki and Alloy as an addition to it. Allowing improvements without interrupting the existing solution.

10. Final Implementation

The monitoring and logging setup was implemented and tested across both Brisbane HQ and Mine Site A. The HQ-MONITOR server is used as the central point for viewing firewall and authentication events through Grafana, while SPAN monitoring is used at Brisbane HQ to allow for a quick view into network traffic. Overall, the monitoring and logging solutions meet the requirements of the project MVP and provide easy visibility for the team to view network and security activity from both network locations.